C9 Chemistry of the Atmosphere – Knowledge Organiser

AQA GCSE Chemistry • Set 1: The Atmosphere Then and Now **Paper 2**

#	Question	Answer
1	Nitrogen in today's atmosphere	About 80%
2	Oxygen in today's atmosphere	About 20%
3	Other gases in small amounts	Carbon dioxide, water vapour, noble gases
4	Source of the early atmosphere	Volcanoes
5	Main gas in the early atmosphere	Carbon dioxide
6	How the oceans formed	Water vapour condensed
7	How CO_2 decreased	Dissolved in oceans; photosynthesis; locked in rocks and fossil fuels
8	Where oxygen came from	Photosynthesis by algae and plants
9	Photosynthesis equation	$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
10	When the atmosphere became like today's	About 200 million years ago

C9 Chemistry of the Atmosphere – Knowledge Organiser

AQA GCSE Chemistry • Set 2: Greenhouse Gases and Climate Change **Paper 2**

#	Question	Answer
1	Main greenhouse gases	Carbon dioxide, methane, water vapour
2	How greenhouse gases warm the Earth	Absorb infrared radiation from the Earth
3	Human sources of carbon dioxide	Burning fossil fuels, deforestation
4	Human sources of methane	Livestock, landfill, rice fields
5	Effects of climate change	Rising sea levels, extreme weather, habitat change
6	Carbon footprint	Total greenhouse gases emitted over a product's life
7	Ways to reduce carbon footprint	Renewable energy, efficiency, carbon capture
8	Why climate predictions are uncertain	Complex systems; simplified models
9	Limestone formed from	Shells and skeletons of sea creatures
10	Crude oil and natural gas formed from	Plankton buried in mud

C9 Chemistry of the Atmosphere – Knowledge Organiser

AQA GCSE Chemistry • Set 3: Atmospheric Pollutants **Paper 2**

#	Question	Answer
1	Source of carbon monoxide	Incomplete combustion
2	Why carbon monoxide is dangerous	Toxic, colourless and odourless
3	Source of sulfur dioxide	Sulfur impurities in fuels
4	Source of oxides of nitrogen	Nitrogen + oxygen at high temperature in engines
5	Problems caused by SO_2 and NO_x	Acid rain, breathing problems
6	Particulates	Solid carbon (soot) and unburnt hydrocarbons
7	Problems caused by particulates	Global dimming, lung damage
8	Global dimming	Less sunlight reaching the Earth's surface
9	Why the early atmosphere is uncertain	Little evidence from 4.6 billion years ago
10	Coal formed from	Plant deposits